WORKED SOLUTION

STEP I 2013, Question 5

Pure • Conics and Rotation

STEP I 2013, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q5

The key idea

The only term that changes with k is the xy term, and it controls what kind of curve we get. With no xy term we can complete the square in x and y separately. When the quadratic part factorises, the whole equation may factorise into two straight lines. And when the quadratic part is a perfect square, $(x+y)^2$, rotating the axes by 45° turns $x+y$ into a single coordinate, leaving a familiar parabola.

Notice too that setting $x=0$ or $y=0$ kills the xy term, so the axis intercepts are the same for every k: $x^2+3x=0$ gives $x=0$ or $x=-3$, and $y^2+y=0$ gives $y=0$ or $y=-1$. Every locus passes through $(0,0)$, $(-3,0)$ and $(0,-1)$.

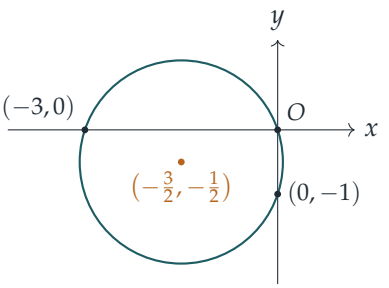
Solution

Part (i)

With $k=0$, completing the square in each variable gives

$$(x+\tfrac{3}{2})^2+(y+\tfrac{1}{2})^2=\tfrac{9}{4}+\tfrac{1}{4}=\tfrac{5}{2}.$$

So the locus is the circle with centre $(-\frac{3}{2},-\frac{1}{2})$ and radius $\sqrt{5/2}\approx 1.58$. It meets the x-axis at $(0,0)$ and $(-3,0)$, and the y-axis at $(0,0)$ and $(0,-1)$. (The origin and $(-3,0)$ are the ends of a chord, so the centre lies on its perpendicular bisector $x=-\frac{3}{2}$, as it should.)



Part (ii)

With $k = \frac{10}{3}$, multiply through by 3:

$$3x^2 + 3y^2 + 10xy + 9x + 3y = 0.$$

The quadratic part factorises as

$$3x^2 + 10xy + 3y^2 = (3x + y)(x + 3y),$$

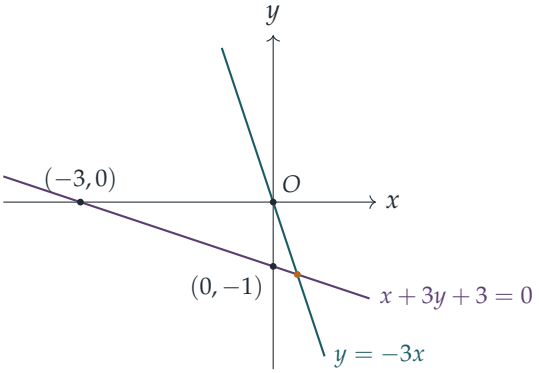
and the linear part is $9x + 3y = 3(3x + y)$, which shares the factor $3x + y$. So

$$(3x + y)(x + 3y) + 3(3x + y) = (3x + y)(x + 3y + 3) = 0.$$

A product is zero exactly when one factor is, so the locus is the pair of straight lines

$$y = -3x \qquad \text{and} \qquad x + 3y + 3 = 0.$$

The first passes through the origin (its only intersection with either axis). The second meets the x-axis at $(-3, 0)$ and the y-axis at $(0, -1)$. Solving the two together, $x - 9x + 3 = 0$, so they cross at $\left(\frac{3}{8}, -\frac{9}{8}\right)$.



Part (iii)

With $k = 2$ the equation is

$$(x + y)^2 + 3x + y = 0.$$

For $\theta = 45°$, $\cos\theta = \sin\theta = \frac{1}{\sqrt{2}}$, so

$$X = \frac{x + y}{\sqrt{2}}, \qquad Y = \frac{y - x}{\sqrt{2}}.$$

Thus $x + y = \sqrt{2}\,X$ and $y - x = \sqrt{2}\,Y$, and adding and subtracting,

$$x = \frac{X - Y}{\sqrt{2}}, \qquad y = \frac{X + Y}{\sqrt{2}}.$$

Ask yourself Why do we need x and y in terms of X and Y, rather than the other way round?

We know P lies on the curve, and we want the equation satisfied by Q. So we must substitute for x and y in the equation of P's locus, which needs them in terms of X and Y. (Because the map is a rotation, the inverse is just the rotation the other way, so this is easy.)

The linear part becomes

$$3x + y = \frac{3X - 3Y + X + Y}{\sqrt{2}} = \frac{4X - 2Y}{\sqrt{2}} = 2\sqrt{2}\,X - \sqrt{2}\,Y,$$

and $(x+y)^2 = 2X^2$. So the locus of Q is

$$2X^2 + 2\sqrt{2}\,X - \sqrt{2}\,Y = 0 \quad\Longleftrightarrow\quad \sqrt{2}\,Y = 2X^2 + 2\sqrt{2}\,X = \left(\sqrt{2}\,X + 1\right)^2 - 1,$$

as required. Every step is reversible, so this is exactly the locus of Q.

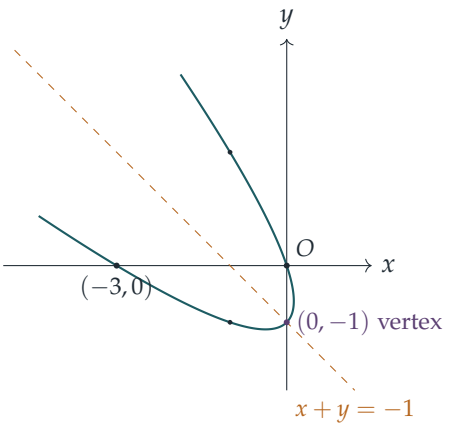
This is a parabola in the (X, Y) plane, opening in the direction of increasing Y. It is symmetric about the line where $\sqrt{2}\,X + 1 = 0$, that is $X = -\frac{1}{\sqrt{2}}$, and its vertex is at $X = -\frac{1}{\sqrt{2}}$, $Y = -\frac{1}{\sqrt{2}}$.

Now we undo the rotation. Q came from P by a clockwise rotation of $45°$, so the locus of P is this parabola rotated $45°$ *anticlockwise*. In terms of x and y:

- the line of symmetry $X = -\frac{1}{\sqrt{2}}$ is $\dfrac{x+y}{\sqrt{2}} = -\dfrac{1}{\sqrt{2}}$, that is
$$x + y = -1;$$
- the vertex is at $x = \dfrac{X - Y}{\sqrt{2}} = 0$, $y = \dfrac{X + Y}{\sqrt{2}} = -1$, that is $(0, -1)$;
- the parabola opens in the direction of increasing $Y = \frac{y-x}{\sqrt{2}}$, which is the direction $(-1, 1)$: up and to the left.

The intercepts are $(0,0)$, $(-3,0)$ and $(0,-1)$, as noted at the start; $(0,-1)$ is the vertex. The tangent there is perpendicular to the line of symmetry, so it has gradient 1: the curve crosses the y-axis at the vertex rather than touching it. As a check on the symmetry, reflecting in $x + y = -1$ sends (a, b) to $(-1-b, -1-a)$, so the origin goes to $(-1,-1)$ and $(-3,0)$ goes to $(-1,2)$; both satisfy $(x+y)^2 + 3x + y = 0$.

For sketching it helps to write $u = x + y$: then $3x + y = -u^2$ and solving gives $x = -\frac{1}{2}(u^2 + u)$, $y = \frac{1}{2}(u^2 + 3u)$.



The locus of P is a parabola with vertex $(0,-1)$ and line of symmetry $x+y=-1$.

Checks

- Substitute the intercepts $(0,0)$, $(-3,0)$, $(0,-1)$ into all three equations: each gives 0, whatever k is.
- In (ii), expand $(3x+y)(x+3y+3)$ and divide by 3 to recover the original equation with $k=\frac{10}{3}$. The crossing point $\left(\frac{3}{8},-\frac{9}{8}\right)$ satisfies both $3x+y=0$ and $x+3y+3=0$.
- In (iii), take $u=1$: the point $(-1,2)$ is on P's locus. Then $X=\frac{1}{\sqrt{2}}$, $Y=\frac{3}{\sqrt{2}}$, and $\sqrt{2}\,Y=3=(1+1)^2-1$.

Common slips

- In (iii), substituting the formulas for X,Y into the equation instead of expressing x,y in terms of X,Y.
- Rotating the parabola back the wrong way, giving the line of symmetry as $y-x=-1$ instead of $x+y=-1$.
- In (ii), missing that the linear terms share the factor $3x+y$, and treating the locus as a conic of some other kind.

Beyond STEP

Every equation $ax^2+bxy+cy^2+dx+ey+f=0$ is a conic, and its type is decided by the discriminant b^2-4ac of the quadratic part: negative gives an ellipse (here a circle, $k=0$), zero a parabola ($k=2$), positive a hyperbola, which can degenerate into a pair of lines ($k=\frac{10}{3}$). In a first-year linear algebra course you learn to remove the xy term systematically by rotating to the eigenvectors of the symmetric matrix of the quadratic form. Strang's *Introduction to Linear Algebra* explains this clearly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.